Data Analysis Essentials
(Common to CSE, IT, AIML & CSE (DS))

Course Code: 2501AI06	L	T	P	C
	0	0	2	2

Course Outcomes:
At the end of the course, student will be able to:

- **CO1** Perform various operations on NumPy arrays.
- **CO2** Importing data from different file formats using pandas
- **CO3** Draw different types of charts using matplotlib.
- **CO4** Demonstrate web scraping using python.
- **CO5** Design preprocessing techniques on loan prediction dataset.

Mapping of Course Outcomes with Program Outcomes:

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	2	2	-	-	2	-	-	1	2	-	-
CO2	3	2	-	-	3	-	-	1	2	-	-
CO3	2	2	-	-	3	-	-	2	1	-	-
CO4	2	2	-	-	1	-	-	1	1	-	-
CO5	2	2	-	-	2	-	-	1	1	-	-

Mapping of Course Outcomes with Program Specific Outcomes:

CO/PSO	PSO1	PSO2
CO1	1	2
CO2	2	-
CO3	2	-
CO4	1	2
CO5	2	-

Practice:
Week 1: Creating a NumPy Array

a. Basic ndarray
b. Array of zeros
c. Array of ones
d. Random numbers in ndarray
e. An array of your choice
f. Imatrix in NumPy
g. Evenly spaced ndarray

Week 2: The Shape and Reshaping of NumPy Array

a. Dimensions of NumPy array

b. Shape of NumPy array
c. Size of NumPy array
d. Reshaping a NumPy array
e. Flattening a NumPy array
f. Transpose of a NumPy array

Week 3: Expanding and Squeezing a NumPy Array

a. Expanding a NumPy array
b. Squeezing a NumPy array
c. Sorting in NumPy Arrays

Week 4: Indexing and Slicing of NumPy Array

a. Slicing 1-D NumPy arrays
b. Slicing 2-D NumPy arrays
c. Slicing 3-D NumPy arrays
d. Negative slicing of NumPy arrays

Week 5: Stacking and Concatenating Numpy Arrays

a. Stacking ndarrays
b. Concatenating ndarrays
c. Broadcasting in Numpy Arrays

Week 6: Perform following operations using pandas

a. Creating dataframe
b. concat()
c. Setting conditions
d. Adding a new column

Week 7: Perform following operations using pandas

a. Filling NaN with string
b. Sorting based on column values
c. groupby()

Week 8: Read the following file formats using pandas

a. Text files
b. CSV files
c. Excel files
d. JSON files

Week 9: Read the following file formats

a. Pickle files
b. Image files using PIL
c. Multiple files using Glob

d. Importing data from database

Week 10: Demonstrate web scraping using python

Week 11: Perform following preprocessing techniques on loan prediction dataset

a. Feature Scaling
b. Feature Standardization
c. Label Encoding
d. One Hot Encoding

Week 12: Perform following visualizations using matplotlib

a. Bar Graph
b. Pie Chart
c. Box Plot
d. Histogram
e. Line Chart and Subplots
f. Scatter Plot.

Additional Practice:

1. Write a program that reads a list of temperatures from a file called temps.txt, converts those temperatures to Fahrenheit, and writes the results to a file called ftemps.txt
2. Write a program to find the greatest number that can be formed by using given set of numbers.
3. Write a program to find sum of digits of a number till you get single digit sum.
   Example:
   Input: 142 (Hint: 1+4+2=7)
   Output: 7
   Input: 4683 (Hint: 4+6+8+3=21 =>2+1=3)
   Output: 3
4. Write a program to count how many times each word present in a file.

Cornerstone Project:

PROJECT 1: Loan Prediction Data Analysis System
Project Implementation, Documentation and Viva Voce

- **Implementation mainly focuses on**
  - Loading and exploring loan dataset using pandas
  - Handling missing values (NaN filling, cleaning)
  - Applying preprocessing techniques:
  - Feature Scaling
  - Standardization
  - Label Encoding & One Hot Encoding
  - Performing exploratory data analysis (EDA)
  - Visualizing insights using matplotlib (bar charts, histograms, etc.)
  - Basic prediction logic using processed data
- **Documentation must contain**
  - Objective & problem definition

- Dataset description (features & target variable)
- Data preprocessing steps explanation
- Visualization results and insights
- Workflow diagram (Data → Preprocess → Analyze → Visualize)
- Screenshots of outputs
- Enhancements (feature selection, correlation analysis)

PROJECT 2: COVID-19 Data Analysis Dashboard

- **Implementation mainly focuses on**
  - Importing CSV/JSON datasets using pandas
  - Cleaning and organizing real-world data
  - Grouping data (country/state-wise analysis using groupby())
  - Time-series analysis of cases
  - Visualization:
  - Line charts (trend analysis)
  - Bar graphs (comparison)
  - Pie charts (distribution)
  - Generating summary statistics
- **Documentation must contain**
  - Objective
  - Dataset source and structure
  - Data cleaning steps
  - Graphical analysis and interpretation
  - Module explanation (Load → Clean → Analyze → Visualize)
  - Screenshots
  - Enhancements (live data scraping)

PROJECT 3: Student Performance Data Analysis System

- **Implementation mainly focuses on**
  - Creating and managing student dataset using pandas
  - Performing operations:
  - Sorting
  - Filtering
  - Adding calculated columns (average, grade)
  - Identifying performance trends
  - Visualization using matplotlib:
  - Bar graph (marks comparison)
  - Pie chart (grade distribution)
  - Generating reports
- **Documentation must contain**
  - Objective
  - Dataset schema
  - Data processing logic
  - Visualization explanation
  - Flowchart of system
  - Screenshots

- Enhancements (subject-wise analysis, topper prediction)

PROJECT 4: Web Scraping and Product Price Analysis System

- **Implementation mainly focuses on**
  - Scraping product data (price, name, rating) using Python
  - Storing scraped data into pandas Data Frame
  - Cleaning and structuring scraped data
  - Performing analysis (price comparison, trends)
  - Visualization:
  - Bar charts (price comparison)
  - Scatter plots (rating vs price)
- **Documentation must contain**
  - Objective
  - Website/data source details
  - Scraping methodology
  - Data cleaning process
  - Analysis and results
  - Screenshots
  - Enhancements (multiple site comparison)

PROJECT 5: Sales Data Analysis System

- **Implementation mainly focuses on**
  - Reading data from CSV/Excel files
  - Data preprocessing and cleaning
  - Performing operations:
  - Groupby (region/product sales)
  - Aggregation (total, average sales)
  - Visualization:
  - Line chart (sales trends)
  - Bar graph (region-wise sales)
  - Histogram (sales distribution)
- **Documentation must contain**
  - Objective
  - Dataset structure
  - Data processing steps
  - Graph interpretation
  - Module flow (Input → Process → Output)
  - Screenshots
  - Enhancements (profit analysis, forecasting basics)

PROJECT 6: File Data Analyzer System

- **Implementation mainly focuses on**
    - Reading multiple file formats:
    - Text, CSV, JSON, Excel
    - Performing operations:
    - Word count
    - Data filtering
    - Statistical analysis
    - Handling multiple files using glob
    - Visualization of extracted data
- **Documentation must contain**
    - Objective
    - File formats handled
    - Processing logic
    - Output results
    - Screenshots
    - Enhancements (automation, batch processing)

PROJECT 7: Social Media Data Analysis System

- **Implementation mainly focuses on**
    - Web scraping social media/text data
    - Cleaning text data
    - Word frequency analysis
    - Visualization:
    - Bar chart (top words)
    - Pie chart (category distribution)
    - Basic sentiment-style categorization
- **Documentation must contain**
    - Objective
    - Data source
    - Preprocessing steps
    - Analysis results
    - Visualization explanation
    - Screenshots
    - Enhancements (real-time scraping, dashboards)

PROJECT 8: Weather Data Analysis System

- **Implementation mainly focuses on**
    - Reading weather dataset (CSV/API/web scraping)
    - Cleaning and preprocessing data
    - Analyzing temperature trends
    - Converting units (Celsius ↔ Fahrenheit)
    - Visualization:
    - Line chart (temperature trends)
    - Histogram (distribution)

- **Documentation must contain**
  - Objective
  - Dataset description
  - Data transformation steps
  - Analysis results
  - Graph interpretation
  - Screenshots
  - Enhancements (forecasting basics)

Text Books:

1. Python Data Science Handbook, Jake VanderPlas, 2nd Edition,2022, ISBN**:** 978-1098121228
2. Data Analysis from Scratch with Python: Step By Step Guide, Peters Morgan,2018, ISBN**:** 978-1725678095

Reference Books:

1. Python for Data Analysis, Wes McKinney, 3rd Edition 2022, ISBN**:** 978-1098104030
2. An Introduction to Statistical Learning: With Applications in Python, Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, Jonathan Taylor, 1st Edition ,2023, ISBN**:** 978-3031387463

Web Links:

1. https://www.analyticsvidhya.com/blog/2020/04/the-ultimate-numpy-tutorial-for-data-science-beginners/
2. https://www.analyticsvidhya.com/blog/2021/07/data-science-with-pandas-2-minutes-guide-to-key-concepts/

Note:

- Students must submit the Certificate of Completion offered by any Industry.
- The capstone project team size shall be four (4) students.
- Students may select any one of the above prescribed projects or a project of their own choice with the prior permission of the Course instructor.
- The SEE – Lab shall be evaluated for 50 marks based on project implementation, oral presentation, 10-minute video presentation, and viva voce.
- The video presentation should consist of the working procedure of the project along with contribution of each student for a minimum of 2 minutes.